Input



Aligned face

224 × 224 × 3

gamma correction + landmark alignment

Feature Extraction

DenseNet-121 backbone (ImageNet-pretrained)

7 × 7 × 1024

Conv block + 2 × Residual block

Conv block + 2 × Residual block

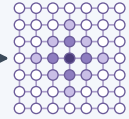
CBAM attention (channel + spatial)

Conv block + 2 × Residual block

Refined feature map 7 × 7 × C

Graph Modeling

Grid-to-graph conversion



49 nodes × C

4-neighbour grid

$\tilde{A} = D^{-1}(A + I)$

node · 1-hop · 2-hop

GraphConv 1 (C → C) + ReLU

GraphConv 2 (C → C) + ReLU

Mean readout over 49 nodes → $g \in \mathbb{R}^C$

GNN-FC (C → C)

Classification

Fully connected + ReLU

Softmax (K = 7)

Class probabilities

